Parameter-Free Attention Network for Spectral–Spatial Hyperspectral Image Classification

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***Abstract*— Hyperspectral images (HSIs) comprise plenty of information in the spatial and spectral domain, which is highly beneficial for performing classification tasks in a very accurate way. Recently, attention mechanisms have been widely used in the HSI classification due to their ability to extract relevant spatial and spectral features. Notwithstanding their positive results, most of the attentional strategies usually introduce a significant number of parameters to be trained, making the models more complex and increasing the computational load. In this article, we develop a new parameter-free attention network for HSI classification. The main advantage of our model is that it does not add parameters to the original network (as opposed to other state-of-the-art approaches) while providing higher classification accuracies. Extensive experimental validations and quantitative comparisons are conducted—using different benchmark HSIs—to illustrate these advantages. The code is available on https://github.com/mhaut/Free2Resnet.**

***Index Terms*— Attention mechanisms, classification, hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL images (HSIs) are collected by hyperspectral imaging spectrometers. Each pixel in these images can be represented by a high-dimensional vector, the entries of which correspond to the spectral reflectance at a specific wavelength. Consequently, each scene contains a wealth of information about the environment and materials captured by the sensor. Considering their rich and detailed spectral and spatial information, HSIs have been successfully applied to many different applications, such as archeology [1], medical analysis [2] environmental monitoring [3], or defense [4], besides others social, industrial, and economic activities. HSI analysis involves a number of processing methods to extract meaningful information from the data, such as unmixing [5], [6], [7], dimensionality reduction and feature extraction [8], [9], [10], or anomaly detection [11], [12], [13], [14].

Over the past few decades, HSI classification has become a hot topic in remote sensing [15], [16]. The goal of classification is to assign pixels to a specific class based on their spectral characteristics. Generally, HSI classification faces two challenges, i.e., the lack of available labeled samples and the large spatial variability of pixel-based spectral features. In this sense, various methods have been developed for HSI classification. Traditional methods were mainly based on pattern recognition, i.e., support vector machines (SVMs) [17], [18], [19] and K-nearest neighbor (KNN) [20]. SVMs show excellent performance when HSIs contain heterogeneous classes with only a few training samples. KNN is often used due to its simplicity in theory and practice. Apart from these methods, the extreme learning machine (ELM) [8], [21], [22], graph-based methods [23], and sparse representation-based classification methods [24] have also been developed to enhance the performance of HSI classification. ELM is mainly utilized to provide probabilistic classification outputs using the extracted features, and it is chosen due to its efficient computation and good classification performance compared with SVMs. Moreover, ELM includes one hidden layer and one linear output layer and thus has a very simple structure. Compared with SVMs, ELM is much faster to train because its input weights are randomly generated and its output weights are analytically computed using a least-squares solution. Graph-based methods usually adopt a low-dimensional subspace to preserve the discriminative information from training samples and exhibit higher stability than other techniques in terms of classification and dimensionality reduction. Sparse representation-based methods usually rely on the observation that hyperspectral pixels can be sparsely represented by linear combinations of a few training samples from a structured dictionary. In particular, the sparse representation of unknown pixels is represented as a sparse vector whose nonzero entries are the selected training samples. The sparse vector is recovered by addressing a sparse-constrained optimization problem, and it can directly determine the class labels of the test samples. However,

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the above methods are based on handcrafted spectral–spatial features, which rely heavily on professional expertise and are very difficult to obtain.

A. Deep Learning: Toward New Classification Horizons

Compared with traditional methods, deep learning has been widely used for HSI classification purposes, receiving increasing attention due to its outstanding representation ability. For instance, autoencoders [25], [26], recurrent neural networks (RNNs) [27], [28], convolutional neural networks (CNNs) [29], [30], [31], [32], [33], morphological CNN (Morph-CNN) [34], and generative adversarial networks (GANs) [35], [36] have been used to explore the high-dimensional complex features of HSIs. Autoencoders learn deep features and store them in hidden layers. If the original input data can be recovered well, this means that such deep features retain enough input information for classification. In [25], a new spectral–spatial feature learning framework was developed for HSI classification where the learned spectral–spatial feature representation is highly discriminative and works very well with linear classifiers, especially when the training data are limited. Ma et al. [26] jointly exploited spectral and spatial features from the HSI's spatial domain to propose an updated deep autoencoder (SDAE) that considers neighbor information in the feature learning step and maximizes the feature difference between classes to enhance the performance of classification. RNN is usually adopted to address the sequence learning problem by adding recurrent edges to connect neurons to themselves across time. Hang et al. [37] proposed a cascaded RNN classification model with gated recurrent units (GRU) that fully exploit the redundant and complementary information of high-dimensional spectral features. Zhou et al. [27] treated each hyperspectral pixel as a sequence of data and used a long short-term memory (LSTM) to model dependencies in the spectral domain to further improve the accuracy of classification. Moreover, CNN is the most popular deep learning model with the ability to automatically extract spectral and spatial features from raw input data via a series of linear transformations (combined with nonlinear activations) to facilitate pattern recognition. In [29], a five-layer 1-D CNN was employed to classify HSIs directly in the spectral domain. This is the first time that CNNs with multiple layers were used for HSI classification purposes. Jia et al. [30] first explored how to build CNNs with HSIs. The proposed method reveals that the performance of classification will be affected by many factors, i.e., the number of iterations and the size of input data, filters, and feature maps, and it is better than state-of-the-art SVM classification methods. However, these methods only consider the spectral information but neglect the spatial information, which is very significant for HSI classification. In this scenario, some deep learning-based methods were developed to incorporate spectral and spatial information to improve the accuracy of classification. Makantasis et al. [31] exploited a CNN to encode the pixels' spectral and spatial information and a multilayer perceptron (MLP) to conduct the classification task. Zhang et al. [32] introduced the dual-channel CNN model into HSI classification, extracted spectral features and spatial features through 1-D CNN and 2-D CNN, respectively, and finally used the softmax regression classifier to combine the resulting spectral and spatial features and provide the final classification results. In [38], a spatial–spectral squeeze-and-excitation (SSSE) structure was developed to adaptively capture the weights for different spectral bands and for different neighboring pixels at the same time. SSSE can suppress or motivate features at certain spectral bands or spatial positions, which can effectively overcome the redundancy in the spectral channels and the pixel inconsistency in the spatial neighborhood. Moreover, several SSSE modules were embedded into a residual network (ResNet) to construct an SSSE-based residual network (SSSERN) model to improve the accuracy of classification.

B. DL Performance Plunges in the Face of HSI

Although existing CNN-based methods often provide good accuracy in classification, they still struggle with HSI due to the high dimensionality of the data, the large number of features, their spectral variability, and the lack of labeled data. Furthermore, CNN-based models present some important shortcomings: 1) they cannot capture the geometric changes of different object regions in HSIs in an adaptive way as they only convolve regular square regions; 2) each convolution kernel has the same weight when convolving all image patches; and 3) they have high computational overloads due to the large number of parameters required. To address these drawbacks, a new multiscale dynamic graph convolutional network (MDGCN) [39] was proposed for HSI classification, and the proposed MDGCN adopted dynamic graphs that are gradually refined during the convolution process.

Recently, attention mechanisms have been developed in language modeling [40], [41], [42] and computer vision tasks [43], [44], [45], achieving promising results on HSI processing [46], [47], [48], [49], [50], [51], [52], [53]. These mechanisms mainly rely on a reasonable assumption that human vision does not process the entire image at once, i.e., it only focuses on a specific part of the entire visual space at "high resolution" and perceives the surrounding environment at "low resolution." Consequently, attention mechanism enables the model to automatically identify the most relevant features and assign higher importance to them, reducing the dimensionality of the data and making the model more computationally efficient. Also, the identification and emphasis of salient features improve the accuracy while providing valuable interpretability and explainability.

Inspired by them, some HSI classification methods based on attention mechanisms have been developed in the literature. In [54], a new end-to-end preactivation residual attention network (PRAN) was developed for HSI classification. PRAN introduces an attention mechanism into the residual block and builds a preactivation residual attention block (PRAB) that can recalibrate the channel feature responses in an adaptive way by explicitly modeling the interdependencies between the channels. Mei et al. [55] developed a spectral–spatial attention network (SSAN), including a spectral attention bidirectional

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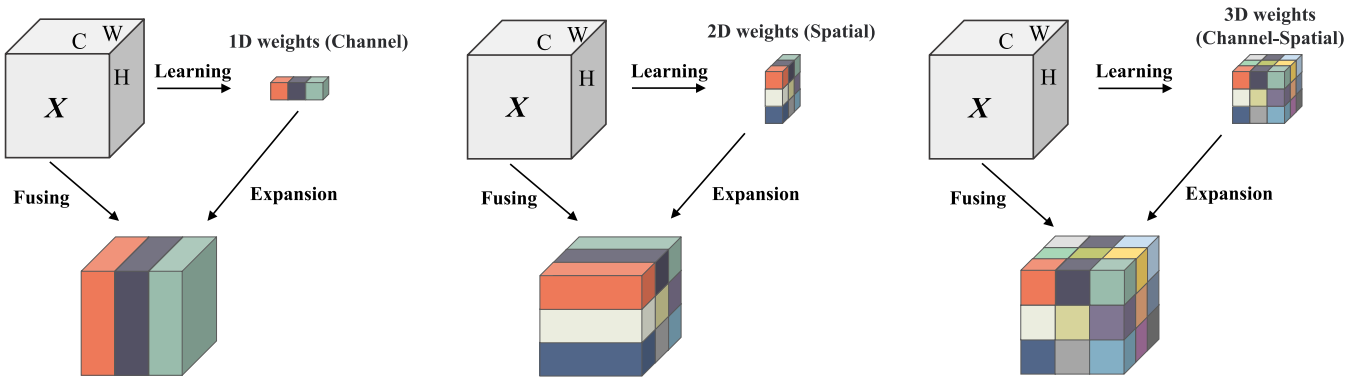


Fig. 1. Illustration of different attention steps. (a) 1-D weights (channel information), which are concerned with the importance of feature channels. The 1-D channel weights are obtained by learning and then fused with the original image for expansion so that the important channels receive more attention during the image processing. (b) 2-D weights (spatial information), which are concerned with spatial importance. Analogously, the 2-D weights guarantee that the important locations obtain more attention during image processing. Obviously, 1-D and 2-D weights are lopsided, and their disaggregated application leads to a loss of spectral–spatial correlation. (c) 3-D weights (channel–spatial information), which focus on both channel and spatial importance, assigning weights to each pixel. Thus, 3-D weighting is able to completely reflect the importance of each pixel in the image, thus capturing the effective information of the image comprehensively.

RNN branch and a spatial attention CNN branch to extract spectral–spatial features for HSI classification. In the proposed SSAN, RNN with attention is used to capture inner spectral correlations within a continuous spectrum, while CNN with attention is designed to focus on saliency features and spatial relevance between neighboring pixels in the spatial dimension. Ma et al. [56] implemented the double-branch multiattention mechanism network (DBMA), which introduces the spectral and spatial branches with channel and a spatial attention module into the fast dense spectral–spatial convolution network (FDSSCN) [57]. In [58], a novel end-to-end 3-D dense convolutional network with spectralwise attention mechanism (MSDN-SA) was proposed for HSI classification. MSDN-SA not only can simultaneously learn spectral and spatial information at different scales by using 3-D dilated convolutions but also introduce a spectralwise attention mechanism to enhance the distinctiveness of spectral features to further improve the accuracy of classification. In [59], another SSAN (sun-SSAN) was proposed to extract discriminative spectral–spatial features from the attention regions of the HSI cube. In particular, the proposed sun-SSAN integrates spectral–spatial feature extraction and attention region extraction into a unified network for end-to-end training, concatenating spectral and spatial modules. Hang et al. [60] proposed a two-branch spectral and spatial attention network for HSI classification. The proposed method incorporates attention modules into each convolutional layer to make CNN focus on more discriminative channels and spatial locations while suppressing unnecessary channels and spatial locations. Yang et al. [61] embedded convolution operations into the transformer structure to develop a new hyperspectral image transformer (HiT) classification network. The proposed HiT method captures the subtle spectral discrepancies and conveys the local spatial context information to enhance the performance of HSI classification. In particular, HiT is the first HSI classification method that employs the 3-D convolution operation to project the input HSIs. Qing et al. [62] integrated 3-D multihead self-attention to propose a 3-D self-attention multiscale feature fusion network (3DSA-MFN) for HSI classification; 3DSA-MFN is the first method that uses convolution kernels of different sizes to learn multiscale feature extraction and adds the feature results extracted from different branches to fuse the spatial and spectral features of the feature map. 3DSA-MFN establishes remote dependency while extracting local features, which can simultaneously improve the local receptive field (RF), learn long-distance interactions, and enhance the performance of HSI classification.

C. All That Glitters Is Not Gold: Challenges of Attention

Notwithstanding the excellent results provided by these attention-based models, the attentional strategies they implement often introduce a significant number of parameters that need to be trained. Indeed, attention mechanisms in DL architectures usually calculate attention scores to determine the importance of different features, by means of a network that takes and processes the input data to extract the scores. This network can be complex and computationally expensive, making the final classification models considerably more complex, compounding the large computational and memory requirements already imposed by deep models. This tradeoff between computation and performance should be carefully considered. Furthermore, such mechanisms are particularly suitable for addressing imbalanced data scenarios, where the quality of newly generated data is critical to achieve the optimal performance [51], [52].

Furthermore, inspired by the CBAM [43], current attention strategies usually decouples the spectral–spatial information contained into the HSI cube [63]. In addition to pure channelwise [49], [58] [Fig. 1(a)] and spatialwise [46] [Fig. 1(b)] attention methods, the literature offers a number of attention mechanisms for HSI classification that use two attention branches or modules, spatial and spectral, to calculate the corresponding scores, which are combined to obtain the desired features [55], [57], [59], [60], [64], [65], [66]. Thus, the close correlation between spectral and spatial dimensions of HSI

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scenes is disaggregated and, consequently, disregarded when computing the attention scores. This can result in the loss of important information and lead to reduced accuracy in the analysis.

D. Contributions

In the end, attention models can drastically increase the computational load, which has a negative impact on model sustainability while neglecting spectral–spatial correlation in the computation of attention scores. In this context, this work provides a solution to straightforwardly obtain 3-D attention scores with a simple structure and low computational load for HSI processing. In particular, a new parameter-free holistic SSAN for HSI classification is presented. First, 3-D scores are obtained without adding complex modules to the network, but from existing features. Thus, it is parameter-free, where the importance of each neuron is inferred directly from its linear separability. The major contributions of our method can be summarized as follows.

1) First and foremost, the proposed approach benefits from attention mechanism to enhance the distinctiveness of spectral and spatial features and further improve the accuracy of classification on 3-D convolution-based architectures.
2) Second, by means of holistic spectral–spatial attention mechanism, the proposed method can adaptively capture the geometric and spectral changes of different object regions in HSIs.
3) Most importantly, the proposed strategy is built on an attention module that derives the scores for the feature map in a layer directly from the linear separability of the neurons rather than adding parameters to the baseline network. As a result, the method uses significantly fewer parameters than other state-of-the-art approaches while yet obtaining higher classification accuracy.

The remainder of this article is organized as follows. Section II introduces the proposed model. Section III presents an experimental evaluation of the proposed method on different datasets. The conclusions of this article and some remarks and hints at plausible future works are given in Section IV.

II. Methodology

In this section, the methodology of the proposed model is introduced. In this context, we provide the general framework through reviewing several existing representative attention modules with a particular focus on the generation of their weights. Then, we describe our proposed attention model and provide a mechanism for solving the attention weights (scores) efficiently. Furthermore, the architecture of our proposed model is introduced for an easy understanding and evaluation of the proposal.

A. Existing Attention Modules

Let us consider $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ as the feature maps obtained from an HSI scene, where C, H, and W denote the number of channels, height, and width of the data, respectively. In this context, the success of convolutional-based networks is based on their automatic feature extraction procedure, which is conducted through the application of multidimensional filters. Thus, a 3-D convolution layer $\mathcal{K} \in \mathbb{R}^{N\times K\times K\times M}$ defines N filters (kernels) that slide according to a stride and padding along the spatial and spectral dimensions of the input data to detect the presence of specific features. To apply it on the HSI data, first, the spectral domain should be rearranged as $\mathbf{X} \in \mathbb{R}^{C\times H\times W\times F}$, where F represents the number of frames. It should be noted that in current programming frameworks, when $F = 1$, it is the same as applying a 2-D convolution, so the values of C and F are often interchanged at the input to apply the 3-D filter, reinterpreting the HSI cube as a collection of 200 frames of a single channel. This enables the 3-D convolution layers to operate the filters between different frames (i.e., original spectral bands), rather than applying them at once to all frames. Indeed, each overlapping of the 3-D kernel applies a linear transformation on $K \times K$ spatial elements along M spectral components of $\mathbf{X}$. The resulting output volume is a set of N channels of $H' \times W' \times F'$ features. As a result, the 3-D kernel operates as a feature detector processing spectral–spatial information in a coupled manner, where the activation function captures the neural responses, introducing some kind of nonlinearities, while pooling techniques provides translation invariance.

Convolution kernels automatically adjust their weights to the data to extract descriptive feature representations exploiting local contextual information, i.e., they extract local patterns, which is the key problem in automatic image processing. In this context, the attention mechanism emerges within network engineering as an approach to fine-grained recognition, with the aim of enriching the feature representation by improving the extraction of distinctive traits. Indeed, this mechanism enhances the weight of some parts of the input data (salient features) while attenuating the weight of other parts (clutter). In this sense, it focuses the attention of the network on a small part of the data that is most important. This is usually conducted through the design and implementation of some computational modules that refine feature maps, conducting a masking procedure. Most commonly, attention mechanisms usually compute a mask $\boldsymbol{\omega}$ by means of several data transformations $\mathcal{G}$ applied on the input, which is normally rearranged across the spectral or spatial domains [46], [49]

$$\begin{aligned} \text{Masking procedure: } \boldsymbol{\omega} &= \text{sofmax}(\mathcal{G}(\mathbf{X}, \boldsymbol{\theta}_\omega)) \\ \text{Attention mechanism: } \mathbf{X}_\omega &= \boldsymbol{\omega} \otimes \mathbf{X}. \end{aligned} \tag{1}$$

As pointed out by (1), in those attentional modules included into end-to-end convolutional models, the mask is automatically fit to the data by a set of trainable parameters $\boldsymbol{\theta}_\omega$ and then thresholded (binarized) by an activation function such as the softmax or the sigmoid. As a result, the attention mask is a gate unit, which is applied to the original input through an elementwise product ($\otimes$) to consolidate the salient features.

One important factor that determines the performance of the attention module is the domain of the attention scores/weights. Existing attention modules are mainly divided into two categories: 1) those that assign attention weights according to channels [see Fig. 1(a)] and 2) those that assign attention weights according to spatial information [see Fig. 1(b)]. The

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resulting weights are 1-D or 2-D. In this sense, most current attention-based processing models usually produce the attention maps separately, computing the channel attention mask on the one hand and the spatial attention mask on the other hand [67]. Nevertheless, 1-D and 2-D weights have limitations in a comprehensive exploration of the image. Obviously, there are inequalities in the importance assigned to either channel or spatial information in the images (the so-called approximation problem) while imposing a significant computational burden and parameter overload. To alleviate the limitations of 1-D and 2-D weights, we utilize 3-D attention weights, which consider both channel and spatial information. As shown in Fig. 1(c), the 3-D attention scores assign weights to both channels and spatial information simultaneously. This improves the ability of convolution-based models to learn more discriminative features.

Another important factor that determines the performance of the attention module is the generation of attention weights. Existing operators for attention weight generation are: spatial average pooling (GAP) or channel average pooling (CAP), spatial max pooling (GMP) or channel max pooling (CMP), batch normalization (BN), and sigmoid. In this regard, Fig. 2 shows the 1-D and 2-D attention weight extraction process. In particular, the following conditions hold.

1) The 1-D attention weights are calculated as follows. First, GAP and GMP are applied to each input feature map to capture global context, getting 1-D vectors. These vectors are fed into a two-layer neural network separately. This two-layer neural network is shared. Finally, the two obtained features are summed and normalized by using the sigmoid to obtain the 1-D weights.
2) The 2-D weights are calculated as follows. Using CAP and CMP on all channels at each location of the feature map, two feature maps are obtained. Then, a convolution operation is performed on these feature maps, which are finally normalized by the sigmoid.

Different from 1-D and 2-D weights, we consider that the way of generating 3-D attention weights should conform to neurological principles. Therefore, we employ a novel approach to generate 3-D weights, as described next.

B. Proposed Attention Module

As mentioned above, existing attention modules focus on either channel or spatial information. Nevertheless, in human vision, individuals pay attention to both sources of information and focus on the important parts of the image. Consequently, we imitate human vision by assigning different weights to channels and spatial information. Therefore, each neuron in the neural network is assigned with additional attention weights. Obviously, how to generate the weights for each neuron becomes a key point.

Existing methods for 3-D weight generation are of two types: 1) encoder–decoder global learning and 2) combination of weights. Both of them have some limitations. For instance, encoder–decoder global learning appends several subnetworks to the neural architecture, which increases the complexity

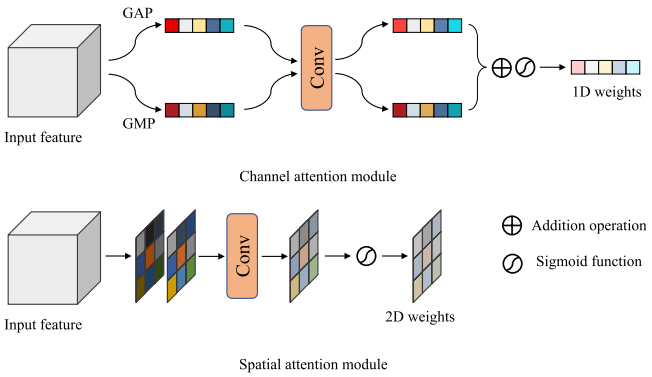


Fig. 2. Generation of 1-D (top) and 2-D (bottom) weights.

of the model and is not really suitable for transfer, while the weight combination method generates the 1-D and 2-D weights separately and then combines them. This approach does not provide an optimal solution for 3-D weights and also increases the computational complexity of the model significantly. In this context, our goal is to provide a solution that straightforwardly obtains 3-D attention weights with a simple structure and low computation load for HS scene processing.

In order to achieve this, we adopt a simple attention module to produce directly 3-D attention weights by estimating the impact of neurons on the resulting feature maps. Different from channelwise and spatialwise attention, the method obtains an attentional weight for each pixel in each channel and spatial position, automatically associating both information at the same time. This is inspired by the visual cortex, where neurons do not respond in isolation to triggering stimuli, but in coordination with neighboring neurons, and although triggering stimuli evoke responses from a large number of neurons, not all contribute equally to perception [68], [69]. In this sense, two main aspects have been considered for measuring the impact of neurons, i.e., their firing patterns and the spatial suppression (SS) effect. Regarding the former, neurons generate a variety of firing patterns to conduct some tasks, with the desired information encoded by their firing rates [70], [71]. In this sense, the most informative neurons tend to be those with the most distinctive firing patterns. Regarding the latter, the visual cortex implements some neural mechanisms that suppress distracting or redundant inputs (i.e., filtering out the irrelevant information that surrounds the target stimulus) to compensate for its own limited processing capacity [72], [73], [74]. Indeed, it has been demonstrated that most neurons have two different regions within their RFs, i.e., the classical RF and the larger surrounding RF. In this context, stimuli located in the classical RF enhance the neural response of a unit, while stimuli located in the surrounding RF indirectly influence the neural response evoked by a stimulus appearing within the classical RF, usually by attenuating its neural response.

In the context of 3-D convolutions, let us interpret each pixel belonging to a feature map as a neuron, i.e., let us assume $\mathbf{X} \in \mathbb{R}^{C\times H\times W\times F}$ as the input feature maps, where C and $P = H \times W \times F$ denote the number of channels and the number of

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neurons on each channel of the data, respectively. Meanwhile, let us denote t as the current target neuron and x_i as other neurons within a particular channel of $\mathbf{X}$, with i indexing the spatial dimensions, i.e., $i \in [1, P)$. For each t and x_i, their corresponding linear transformations $\hat{t}$ and $\hat{x}_i$ are obtained as follows:

$$\begin{aligned} \hat{t} &= w_t t + b_t \\ \hat{x}_i &= w_t x_i + b_t \end{aligned} \tag{2}$$

where w_t and b_t denote the corresponding weight and bias of the neuron, respectively.

Then, guided by the neurological fact that important neurons transmit specific impulses to their environment and inhibit the activity of surrounding neurons, the importance of each neuron is obtained as its linear separability relative to other neurons. In particular, the linear separability of the target neuron t and its surrounding neurons x_i is obtained as follows:

$$e_t\left(w_t, b_t, t^*, x^*, x_i\right) = \left(t^* - \hat{t}\right)^2 + \frac{1}{P-1}\sum_{i=1}^{P-1}\left(x^* - \hat{x}_i\right)^2. \tag{3}$$

From (3), it can be seen that, by minimizing the energy function $e_t(w_t, b_t, t^*, x^*, x_i)$, the current neuron can be linearly separated from all other neurons, i.e., the smaller $e_t(w_t, b_t, t^*, x^*, x_i)$, the more different the target neuron t is from the surrounding neurons, and thus, the more important t is. Furthermore, the energy function achieves a minimum value when $t^* = \hat{t}$, $x^* = \hat{x}$ and $t^* \neq x^*$. To facilitate calculations, we let t^* and x^* be binary values, i.e., $t^* = 1$ and $x^* = -1$, and we replace $\hat{t}$ and $\hat{x}$ by their original expressions in (2), thus (3) can be expressed as

$$\begin{aligned} &e_t(w_t, b_t, 1, -1, x_i) \\ &\quad = \left(1 - \hat{t}\right)^2 + \frac{1}{P-1}\sum_{i=1}^{P-1}\left(-1 - \hat{x}_i\right)^2 \\ &\quad = (1 - (w_t t + b_t)) + \frac{1}{P-1}\sum_{i=1}^{P-1}(-1 - (w_t x_i + b_t))^2. \end{aligned}$$

In addition, a regularizer term controlled by the hyperparameter $\lambda = 1\mathrm{e}^{-4}$ can be included in the final energy function of each channel as follows:

$$\begin{aligned} &e_t(w_t, b_t, 1, -1, x_i) \\ &= (1-(w_t t + b_t))^2 + \frac{1}{P-1}\sum_{i=1}^{P-1}(-1 - (w_t x_i + b_t))^2 + \lambda {w_t}^2. \end{aligned} \tag{4}$$

Solving (4) through traditional iterative solution methods—such as stochastic gradient descent (SGD)—is significantly time-consuming as P functions must be solved for each channel. Fortunately, through the partial derivatives of the energy function $e_t(w_t, b_t, t^*, x^*, x_i)$ with respect to the weights w_t and biases b_t, we can obtain the solution of w_t and b_t as follows:

$$\begin{aligned} w_t &= -\frac{2(t - \mu_t)}{(t - \mu_t)^2 + 2\sigma_t^2 + 2\lambda} \\ b_t &= -\frac{1}{2}(t - \mu_t) w_t \end{aligned} \tag{5}$$

where $\mu_t = 1/(P-1)\sum_{i=1}^{P-1} x_i$ and $\sigma_t^2 = 1/(P-1)\sum_{i=1}^{P-1}(x_i - \mu_t)^2$ denote the mean and variance obtained from all neurons except the current neuron t, respectively. Under the assumption that all channel neurons are independent and equally distributed, we can calculate the mean and variance of all neurons as $\mu = (1/P)\sum_{i=1}^{P} x_i$ and $\sigma^2 = (1/P)\sum_{i=1}^{P}(x_i - \mu)^2$, reducing the computational complexity of the model. Therefore, the minimal energy of all neurons can be obtained as follows:

$$e_t^* = \frac{4\left(\sigma^2 + \lambda\right)}{(t - \mu)^2 + 2\sigma^2 + 2\lambda}. \tag{6}$$

As the lower the energy e_t^*, the more the current neuron is distinguished from the surrounding neurons, the importance of the current neuron can be expressed by $1/e_t^*$. Besides, since neural attention mechanisms are expressed by scaling signals, a scaling operator is used to perform feature refinement as follows:

$$\mathbf{X}_\omega = \mathrm{sigmoid}\left(\frac{1}{\mathbf{E}}\right) \otimes \mathbf{X} \tag{7}$$

where $\mathbf{E}$ integrates all e_t^* across channel and spatial dimensions. The sigmoid function is used to force $1/\mathbf{E}$ into the range [0, 1] and does not impact the relative importance between all neurons. Hereinafter, we refer to our model as `Free`^{2}`-ResNet`.

C. Network Architecture

The network architecture is based on the ResNet, one of the most widely used variations of the CNN [75], [76]. In particular, ResNet improves the expression and learning ability of the CNN model by deepening the network. It is noteworthy that, as the depth of the model increases, a gradient vanishing problem appears that slows down the convergence speed of the network. To overcome this drawback, the original ResNet combined the residual representation concept with a CNN model to form a block structure with basic residual learning. The core idea of this structure is the superposition of the residual function mapping layer based on a shallow layer network, which can conduct residual learning to form skip connections, solving the problem of gradient disappearance, and further improving the accuracy of feature extraction [77]. features The graphical visualization of the proposed network for spectral–spatial HSI classification is shown in Fig. 3. The network comprises three main parts: 1) the convolution-based head; 2) the feature-extractor body; and 3) the classification tail. Regarding the former, the network head is inspired by inception modules [78] and depthwise and pointwise convolutions [79] to extract the spatial and channel information from input hyperspectral scenes. As a result, the input feeds two branches, the first one comprising 24 filters of size $1 \times 1 \times 7$ and the second one comprising 24 filters of size $3 \times 3 \times 7$. The output volumes contain 24 channels of several frames each one. To get the final volume, the obtained outputs are combined in the channel dimension by a sum of frames within the corresponding channel, merging the previously extracted spectral–spatial features.

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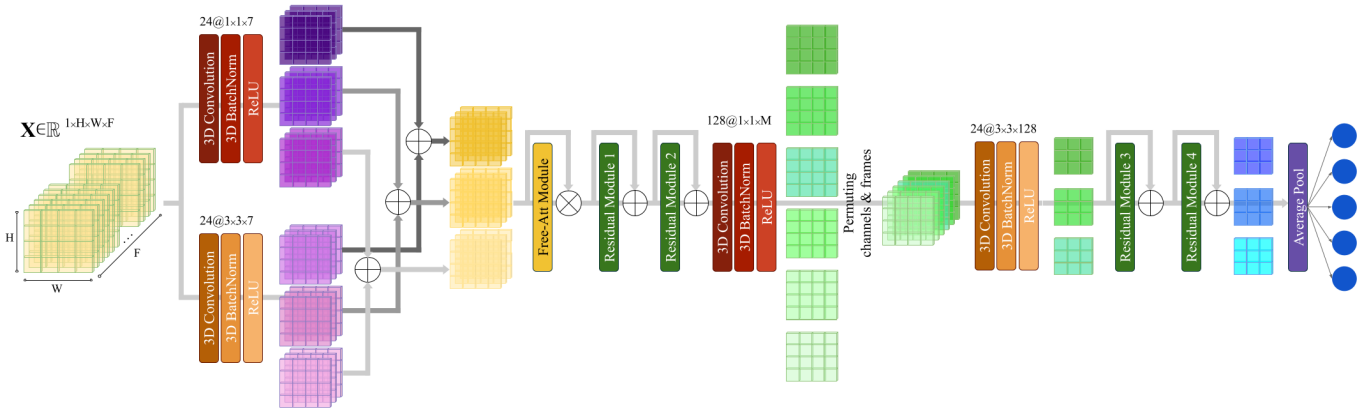


Fig. 3. Proposed network architecture (M indicates that the spectral dimension of the kernel is adapted to the input data to obtain N channels with 1 frame/channel).

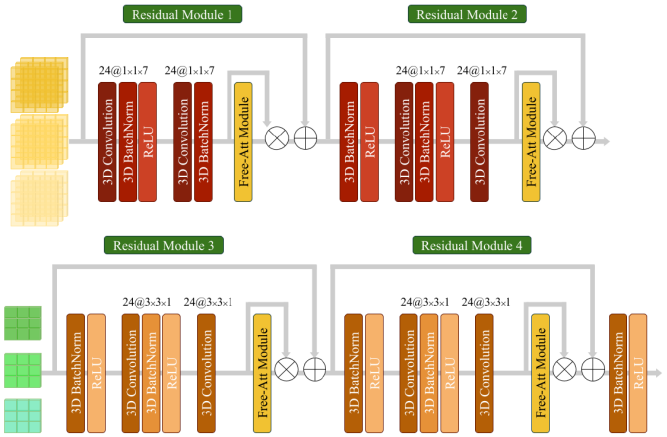


Fig. 4. Graphical detail of residual modules 1 and 2 (top row) comprising pointwise convolutions with 1×1 kernels, and residual modules 3 and 4 (bottom row) comprising spatial convolutions with 3×3 kernels.

This resulting volume feeds into the body of the proposed network, which proceeds by applying the parameter-free attention mechanism. Indeed, from the input data, the parameter-free attention module first extracts the energy values of the input given by (3), thus creating the attention mask, which is applied over the input volume to highlight those features that most affect classification, just following (7). Two residual modules with pointwise convolutions are then applied (see the top row of Fig. 4 for further details). These modules combine information along the frames with a low computational cost, reducing the effects of data degradation and vanishing gradient through the implementation of the identity function in the residual connection while applying at the end of the module the attention mechanism to enhance the extraction of relevant features. After pointwise-based residual modules, a final pointwise convolution with 128 filters of size $1 \times 1 \times M$ is applied, where M is adapted to the number of frames per channel. In fact, this convolution combines the frames of each channel into a single one, increasing the number of output channels. This improves the feature representation of the data along the spectral dimension, refining the extracted features, as well as considerably reducing the size of the output volume. Then, the data are reshaped, exchanging channels for frames to apply a convolution with 24 filters of $3 \times 3 \times 128$, which combines the spatial and spectral information contained into the input volume. Each application takes all the frames of the channel, recombining the features. At this point, two depthwise residual modules, with convolutions of size $3 \times 3 \times 1$, are used to combine the spatial information, requiring a low computational effort. Also, feature extraction is enhanced by the free-parameter attention mechanism. The final features are processed by an average pool to flatten them, in order to apply the final classification layer, which is implemented by a standard fully connected layer.

It can be seen that the design of the model is clearly determined by limiting floating-point operation through the application of pointwise and depthwise lightweight kernels while enhancing the feature representation with the proposed parameter-free attention mechanism for spectral–spatial hyperspecral image classification.

III. Experimental Results

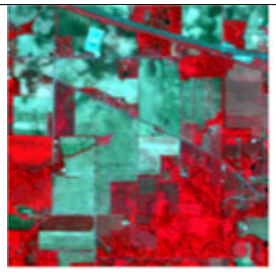
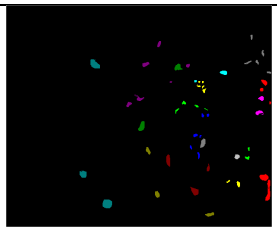
A. Hyperspectral Datasets

In our experiments, three widely used benchmarks for hyperspectral classification are adopted: Indian Pines (IP), Kennedy Space Center (KSC), and University of Pavia (UP). The pseudo-color images and ground truth maps of the three datasets are shown in Table I. The IP and KSC datasets mainly include natural types of land covers, while the UP dataset is composed of both natural and man-made types of land covers. The detailed description is given as follows.

1) *IP:* The IP dataset (Table I) was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [80] on June 12, 1992, covering the area of the Indian Creek and Pine Creek watersheds. The selected image includes 145×145 pixels with 200 bands and 20-m spatial resolution. As shown in Table I, this dataset is composed of 16 types of land cover plants and the number of samples of different classes is seriously

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TABLE I
PSEUDO-COLOR IMAGES AND GROUND TRUTH MAPS OF IP, KSC, AND UP DATASETS

Data sets	pseudo-color image	ground truth map	Label
IP			Alfalfa, Corn-n, Corn-m, Corn, Grass-p, Grass-t, Grass-m, Hay-win, Oats, Soybean-n, Soybean-m, Soybean-c, Wheat, Woods, Buildings, Stone
KSC			Unlabelled, Scrub, Willow, Palm, Pine, Broadleaf, Hardwood, Swap, Graminoid, Spartina, Cattail, Salt, Mud, Water
UP			Unlabelled, Asphalt, Meadows, Gravel, Trees, Metal, Bare Soil, Bitumen, Bricks, Shadows

unbalanced. Due to the similarity among all categories, it is challenging to classify this dataset accurately.

2) *KSC:* The KSC dataset (Table I) was also acquired by the AVIRIS sensor over the KSC in FL, USA, on March 23, 1996. After removing the low information bands, the image has 176 bands in the wavelength range of 400–2500 nm. An image with 512 × 614 pixels and 18-m spatial resolution is considered, as shown in Table I. This dataset contains 13 types of land covers, where the samples and the definition of classes were collected by KSC personnel according to the land cover maps resulting from color infrared photographies and Landsat Thematic Mapper (TM) imagery.
3) *UP:* The UP dataset was obtained by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor [81] over Pavia, Northern Italy. Different from the first two datasets, UP is an urban scene with a high spatial resolution of 1.3 m, and it contains 610 × 340 pixels and 103 bands in the wavelength range of 0.43–0.86 μm after removing the noisy bands. From Table I, it can be seen that nine types of land covers are present in this image (asphalt, meadows, gravel, trees, metal, bare soil, bitumen, bricks, and shadows). Intuitively, the differences among the categories of UP dataset are larger than that in the other two datasets.

TABLE II
OBTAINED RESULTS FOR ALTERNATIVE 3DCNN ARCHITECTURE. FOR THREE KEY DATASETS, NAMELY, IP, UP, AND KSC, ACCURACY (%) PER CLASS, OA, AA, KAPPA, AND PARAMETERS ARE SHOWN

Class	Indian Pines		University of Pavia		Kennedy Space Center	
	3DCNN	A3DCNN	3DCNN	A3DCNN	3DCNN	A3DCNN
0	61.50±7.18	**67.00**±6.40	**99.73**±0.10	99.70±0.10	99.06±0.19	**99.29**±0.36
1	90.77±1.78	**92.81**±1.27	99.88±0.06	**99.93**±0.04	95.24±2.97	**96.83**±1.38
2	92.65±1.94	**95.64**±1.74	**97.33**±0.52	97.11±0.47	83.87±5.60	**91.42**±3.92
3	79.90±9.28	**82.17**±6.69	99.14±0.29	**99.48**±0.21	65.51±16.45	**78.45**±3.36
4	91.33±1.06	**92.11**±1.13	**100.0**±0.00	100.0±0.00	50.08±31.35	**74.73**±3.65
5	**98.26**±0.60	96.22±1.28	99.75±0.06	**99.88**±0.09	70.11±22.19	**86.02**±3.85
6	**80.00**±6.16	70.91±6.80	98.52±0.40	**98.92**±0.30	89.52±8.70	**95.00**±3.94
7	97.94±1.20	**98.31**±1.75	99.39±0.27	**99.56**±0.12	94.18±3.15	**96.44**±1.40
8	47.50±22.91	**48.75**±24.81	**100.0**±0.00	100.0±0.00	98.97±0.62	**99.72**±0.18
9	93.89±1.09	**94.82**±0.78	-	-	96.47±1.24	**98.14**±0.96
10	93.62±1.09	**95.21**±0.34	-	-	**100.0**±0.00	100.0±0.00
11	89.61±1.75	**91.48**±3.55	-	-	99.31±1.15	**99.75**±0.38
12	86.55±2.95	**91.67**±5.04	-	-	**100.0**±0.00	100.0±0.00
13	**98.49**±0.44	98.33±0.65	-	-	-	-
14	89.51±4.51	**89.77**±2.51	-	-	-	-
15	**98.73**±1.39	97.97±1.29	-	-	-	-
OA	93.12±0.75	**94.27**±0.68	99.58±0.04	**99.66**±0.05	93.17±2.43	**96.29**±0.37
AA	86.89±1.62	**87.70**±0.82	99.31±0.12	**99.40**±0.06	87.87±5.17	**93.52**±0.62
Kappa(x100)	92.15±0.85	**93.46**±0.77	99.45±0.05	**99.55**±0.06	92.40±2.71	**95.87**±0.41
Parameters	1516492		573585		1285189	

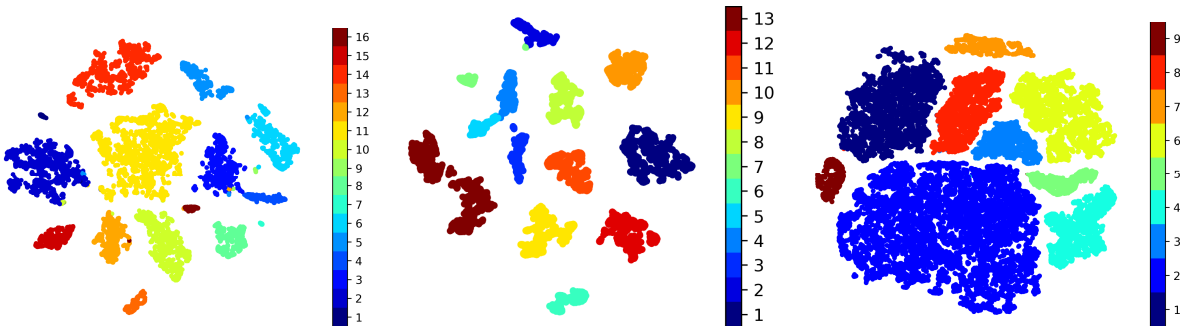
B. Experimental Settings

To test the performance of the proposed parameter-free attention module, different thematic experiment have been

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TABLE III
ACCURACY RESULTS IN TERMS OF OA, AA, AND KAPPA FOR DIFFERENT ATTENTION MODULES WITHIN THE PROPOSED FRAMEWORK

Dataset	Metric	Baseline	Local Attention	Global Attention	Free^2-ResNet
IP	OA	99.01±0.3333	99.03±0.2338	99.07±0.2874	**99.28**±0.2705
	AA	96.27±1.7340	96.20±2.6617	96.30±2.0277	**97.04**±1.8076
	Kappa	0.9887±0.0038	0.9889±0.0026	0.9894±0.0033	**0.9918**±0.0031
KSC	OA	99.34±0.2399	99.40±0.2407	99.41±0.2682	**99.62**±0.1684
	AA	98.91±0.4737	98.88±0.632	99.01±0.5486	**99.35**±0.3226
	Kappa	0.9927±0.0027	0.9934±0.0027	0.9934±0.003	**0.9957**±0.0019
UP	OA	99.24±0.1297	99.30±0.10	99.31±0.1157	**99.32**±0.1534
	AA	98.69±0.1759	98.80±0.1782	98.80±0.1725	**98.81**±0.2545
	Kappa	0.9899±0.0017	0.9908±0.0013	0.9909±0.0015	**0.991**±0.002



(a) (b) (c)

Fig. 5. Visualization of the proposed spectral–spatial features for the test samples in different datasets via t-SNE. Points represent the features of test samples and their class labels are shown with different colors: (a) IP, (b) KSC, and (c) UP (best viewed in color).

conducted, the former to analyze the benefits of the method on different network architectures and through ablative studies and the latter to evaluate the performance of the proposed classification. The widely used overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) metrics have been considered to quantitatively assess the classification performance of the method. Furthermore, we provide the runtime and number of parameters to evaluate the complexity of the proposed classifier.

The first set of experiments evaluates the benefits of the parameter-free attention strategy on different architectures, i.e., a standard 3DCNN network and a ResNet network, with the aim of testing the usability and portability of the module on different neural architectures. The obtained results in terms of OA, AA, Kappa coefficient, and number of parameters are provided in Table II. Furthermore, considering the residual model, hereafter Free^2-ResNet, a comprehensive ablative study has been carried out, in order to analyze the impact of each attention block included in the architecture proposed in Fig. 3. Table III presents the results in terms of OA, AA, and Kappa coefficient. Finally, considering Free^2-ResNet, the class separability in the considered spectral–spatial feature space is discussed via t-SNE. The visualization result is shown in Fig. 5, in which every class was represented by a unique color.

Regarding the second set of experiments, to test the performance of the proposed Free^2-ResNet for HSI classification, five classical models (i.e., RF [88], multinomial logistic regression (MLR) [89], GRU [90], SVM [91], and LSTM [28]) and 11 convolution-based neural networks for classification (i.e., ContextNet [82], DPyResNet [76], ResNet [75], SSRN [83], MS-3DNet [84], ENL-FCN [85], A^{2}S^{2}K-ResNet [48], three transformer networks—i.e., ViT [86], LeViT [87], and HiT [61]—, and SSAN [55]) are taken to draw a comparison between different classification methods. It should be noted that the classical models are mainly spectral classifiers, while the considered deep neural network classifiers are spectral–spatial classifiers (as the proposed Free^2-ResNet). Also, it is worth noting that the proposal is compared with five contemporary methods, i.e., A^{2}S^{2}K-ResNet and SSAN and the transformer networks ViT, LeViT, and HiT. For the sake of fairness, only 10% of training samples for both classical and deep neural networks classifiers are used to classify the three datasets IP, KSC, and UP. Regarding the transformers, they have been compared following the original work of [61], with 10% of IP and UP using $15 \times 15 \times C$ window size. The SSAN has also been compared over IP and UP, using 10% of labeled data. Furthermore, the original work of [55] considers the input patches of $24 \times 24 \times C$. To conduct a fair comparison, the window size has been adjusted to $9 \times 9 \times C$. The classification accuracy of each class as well as the OA, AA, and κ values of all methods for the three datasets are computed and shown in Tables IV–VI. Among them, the best result is marked in bold. In general, the classification performance of deep neural networks outperforms that of the classical methods. Therefore, only the classification maps of deep neural network classifiers are shown in Figs. 6–8 for simplicity. Also, the number of parameters and the running time of the deep neural network models, including the proposed Free^2-ResNet, are listed in Table VII.

The last set of experiments conducts a discussion about the impact of different factors in the classification performance and complexity of the proposed network, comparing the best DL-based classifiers of the previous classification experiments.

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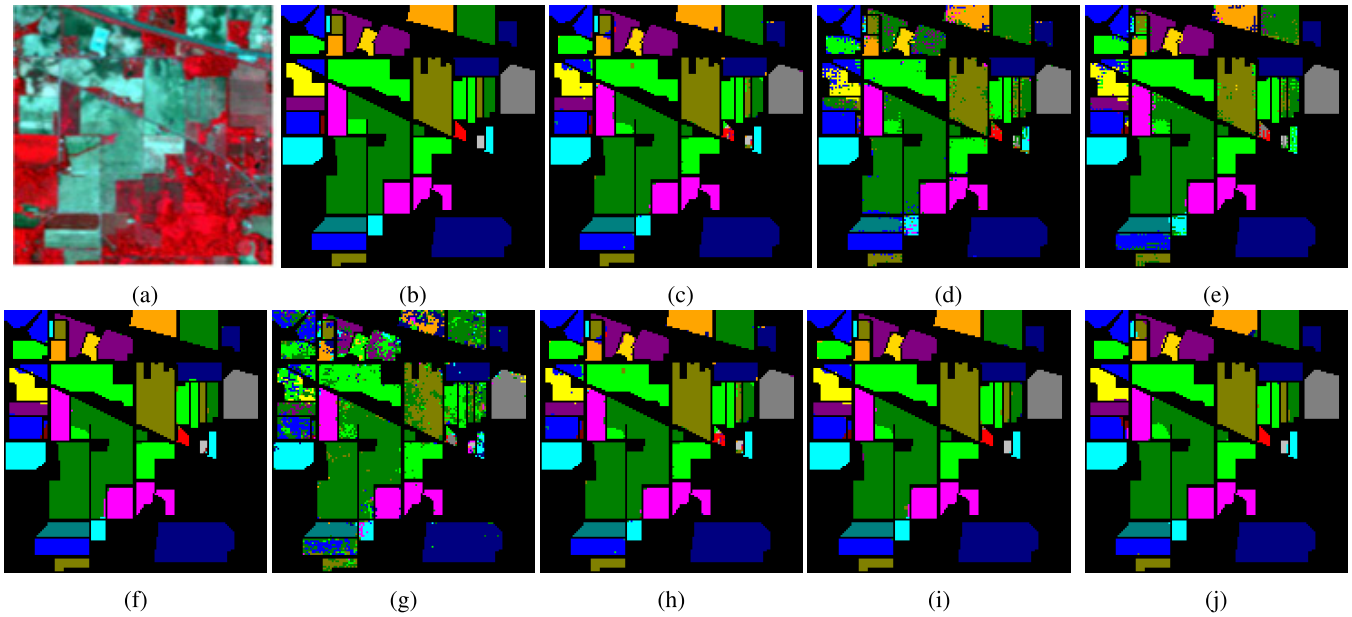


Fig. 6. Classification maps for the IP dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by ContextNet, DPyResNet, ResNet, SSRN, MS-3DNet, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed Free^2-ResNet, respectively.

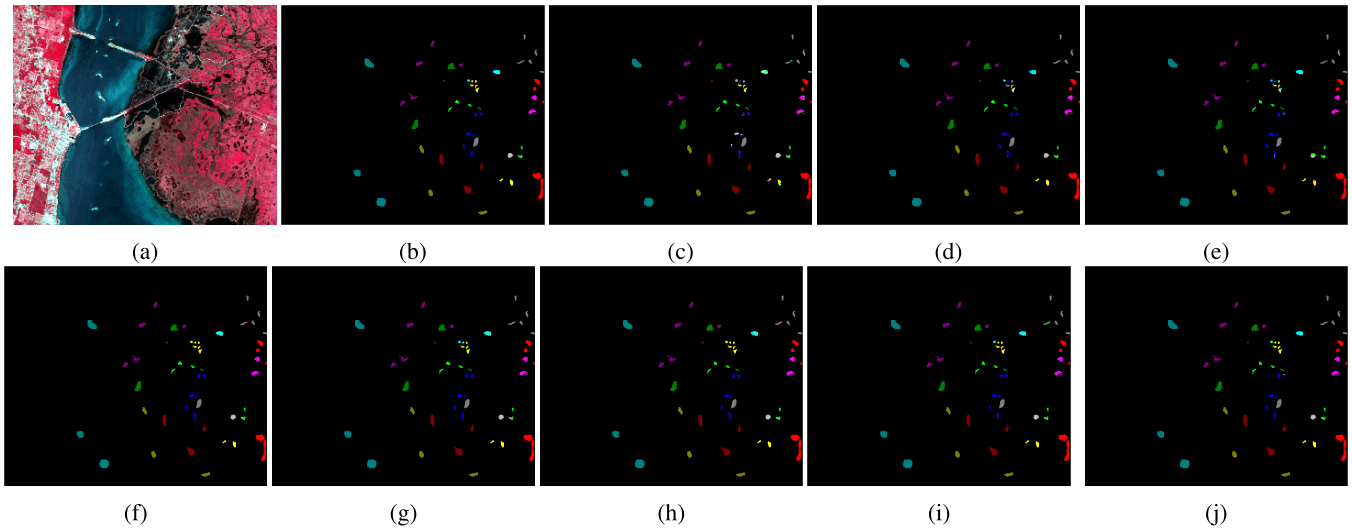


Fig. 7. Classification maps for the KSC dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by ContextNet, DPyResNet, ResNet, SSRN, MS-3DNet, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed Free^2-ResNet, respectively.

In particular, the first discussion topic is the OA as a function of different spatial window sizes. Among the deep neural network classifiers, including the proposed Free^2-ResNet, all of them (with the exception of ContextNet) use a spatial window to capture spatial information. Table VIII shows the OA of the other deep neural network methods considering different spatial window sizes (7×7, 9×9, and 11×11) and 10% of samples for IP, KSC, and UP datasets. It should be mentioned that the spatial window size resulting in optimal performance of the proposed Free^2-ResNet has been included in the setting range. The second discussion topic is the OA as a function of different training sample sizes. In our experiments, 2%, 3%, 5%, 10%, 15%, and 20% of samples are randomly extracted from each class to test the classification performance of all the deep neural network classifiers and the proposed Free^2-ResNet. Fig. 9 shows the OA of the considered methods as a function of the number of training samples for the IP, KSC, and UP datasets Finally, the last topic of interest analyzes the effect of class imbalance scenarios on the integration of attention modules into the model architecture. Concretely, the comparison is conducted against D-SSN and S3Net [53] using the first 60 principal components from the principal component analysis (PCA) and the patch size of 13×13.

C. Benefits of Parameter-Free Attention Strategy

1) Improvements Over Standard 3DCNN Architectures: The parameter-free attention strategy can be easily included in different deep architectures. In this sense, we have tested the benefits of the attention mechanism on a standard 3DCNN architecture. The implementation of the 3DCNNs is simple and straightforward. This classifier comprises a feature

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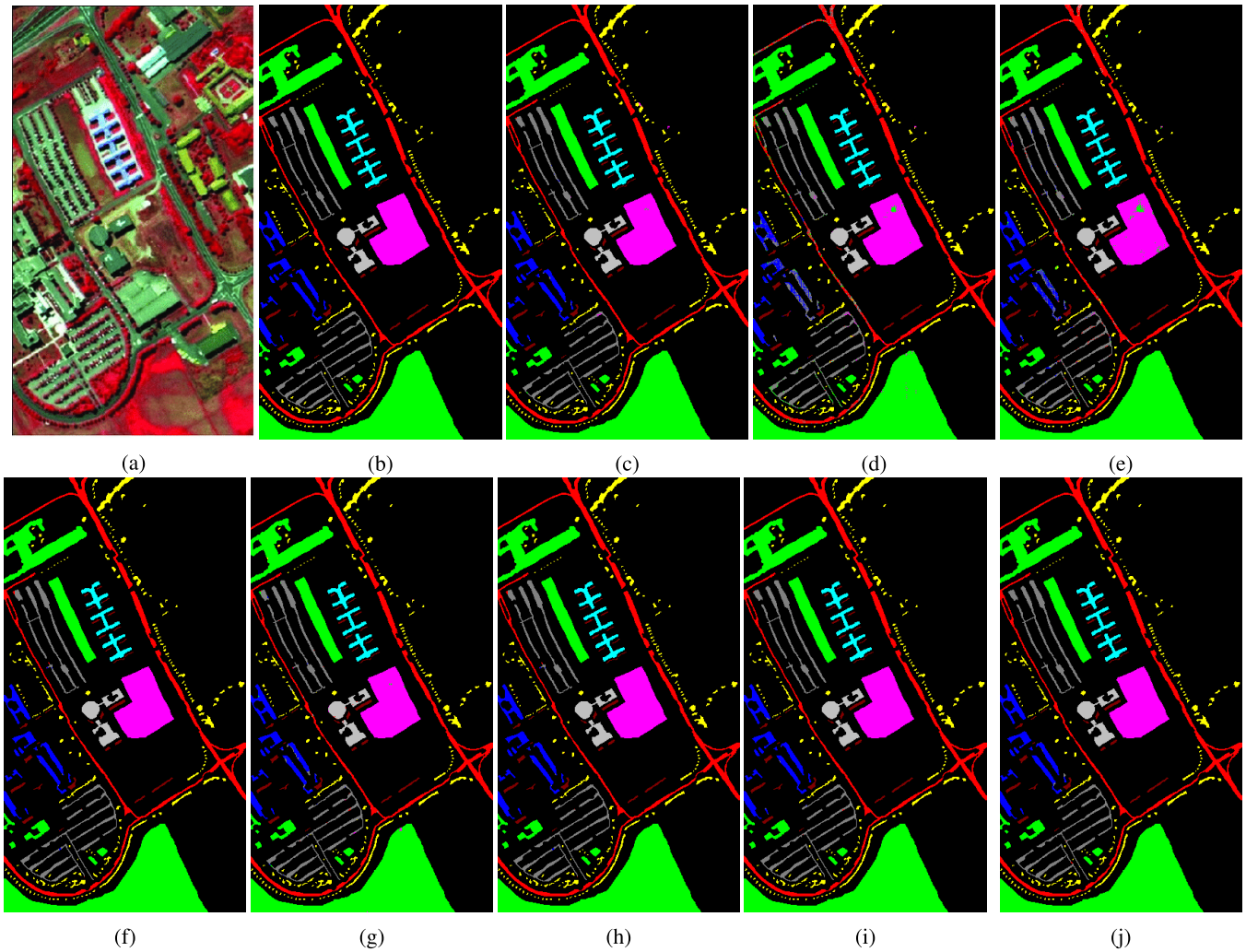


Fig. 8. Classification maps for the UP dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by ContextNet, DPyResNet, ResNet, SSRN, MS-3DNet, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed Free^2-ResNet, respectively.

extractor with three convolution layers ($3 \times 3 \times 24$@16, $3 \times 3 \times 16$@32, and $3 \times 3 \times 8$@64) followed by their respective BN and ReLU functions, and an MLP classifier with two fully connected layers of 300 and N_{classes} neurons. The attention-based network (namely, A3DCNN) introduces the parameter-free attention module after each convolution layer, refining the extracted feature maps. Table II provides the obtained classification results in terms of accuracy per class, OA, AA, and kappa coefficient. In addition, the number of parameters is supplied to analyze the impact of the attention mechanism on the model complexity. It must be noted that the results have been extracted on IP, UP, and KSC scenes using 10% of labeled samples to conduct the training.

Regarding to the excellent results obtained over IP, the categorization of most classes is remarkably improved in contrast to the standard 3DCNN. A similar behavior is observed for the other classes where the standard method is superior to the A3DCNN. On the other hand, due to the extremely high precision values for all classes, the classification accuracy differences for both methods over UP are practically negligible. This is largely due to the absence of minority classes. It is noteworthy that the A3DCNN proposal also manages to overcome the standard model for all the global values of OA, AA, and kappa, which is quite complex due to the difficulty of obtaining gains with such high accuracy. Last but not least, the improvement in the KSC scene is evident. Due to the high number of unlabelled samples, the complexity of the dataset is high, as observed by the standard deviations of the 3DCNN. On the contrary, the proposal succeeds in exploiting the attention method where the detection of the samples among background noise becomes more efficient. Thus, the standard deviation is reduced and the accuracy is increased. In terms of complexity, the number of parameters is observed to be equal in both methods, benefiting from zero added complexity.

2) Ablation Study: To conduct a comprehensive analysis and evaluation of the proposed model, an ablation study has been performed for all datasets to draw a comparison between different attention modules that can be implemented and included in the network backbone. In this context, different implementations have been tested: the baseline shares the same architecture that the proposed Free^2-ResNet but without

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TABLE IV

OA, AA, AND κ VALUES ON THE IP DATASET USING A FIXED SET OF 10% OF THE AVAILABLE LABELED SAMPLES (BOLD INDICATES THE BEST RESULTS)

Class	Classical Models					Deep Neural Networks										
	RF [88]	MLR [89]	GRU [90]	SVM [91]	LSTM [28]	ContextNet [82]	DPyResNet [76]	ResNet [75]	SSRN [83]	MS-3DNet [84]	ENL-FCN [85]	A^{2}S^{2}K-ResNet [48]	ViT [86]	LeViT [87]	HiT [58]	SSAN [55]	Free^2-ResNet
1	28.46 ±0.06	15.45 ±0.02	69.92 ±0.14	51.22 ±0.19	69.11 ±0.09	88.78 ±0.08	94.59 ±0.08	98.66 ±0.02	57.78 ±0.42	66.67 ±0.47	97.56 ±0.00	97.56 ±0.03	48.48	68.42	94.25	45.68	**99.55**±1.01
2	56.63 ±0.02	73.77 ±0.01	76.96 ±0.01	81.22 ±0.04	74.22 ±0.02	98.19 ±0.01	93.83 ±0.04	87.85 ±0.03	98.37 ±0.01	75.94 ±0.08	93.15 ±0.00	98.62 ±0.01	66.69	68.15	92.68	53.30	**99.19**±0.36
3	48.42 ±0.01	51.14 ±0.03	67.20 ±0.04	65.82 ±0.01	71.49 ±0.03	95.37 ±0.03	89.30 ±0.01	92.71 ±0.01	97.47 ±0.01	81.39 ±0.01	**97.59**±0.00	98.58 ±0.01	51.60	59.75	78.55	53.20	98.46±0.56
4	33.49 ±0.03	43.97 ±0.05	61.82 ±0.04	57.75 ±0.04	60.72 ±0.04	97.04 ±0.02	93.51 ±0.06	95.43 ±0.05	**99.12** ±0.01	88.63 ±0.06	91.55 ±0.00	98.29 ±0.01	56.66	72.73	86.73	34.18	97.66±3.78
5	85.21 ±0.03	83.52 ±0.03	85.36 ±0.02	90.04 ±0.01	87.51 ±0.02	97.78 ±0.02	**99.26**±0.01	98.23 ±0.02	97.79 ±0.01	95.61 ±0.05	97.47 ±0.00	99.02 ±0.01	47.81	44.07	85.53	78.23	99.87±0.20
6	92.64 ±0.03	94.82 ±0.01	94.22 ±0.01	96.25 ±0.01	94.77 ±0.01	98.60 ±0.01	98.52 ±0.01	97.98 ±0.01	98.50 ±0.01	96.78 ±0.03	99.24 ±0.00	98.71 ±0.01	77.00	85.80	98.32	91.73	**99.66**±0.38
7	2.67 ±0.04	41.33 ±0.19	50.67 ±0.07	73.33 ±0.02	85.33 ±0.09	90.35 ±0.10	83.08 ±0.18	92.98 ±0.10	66.67 ±0.47	100.00 ±0.00	**100.00**±0.00	93.10 ±0.10	40.00	14.63	92.00	10.42	90.91±12.03
8	97.67 ±0.02	98.53 ±0.01	97.83 ±0.00	97.98 ±0.01	97.83 ±0.01	97.76 ±0.03	97.63 ±0.02	95.06 ±0.01	96.45 ±0.03	89.51 ±0.09	97.44 ±0.00	98.83 ±0.02	87.32	89.33	94.63	99.49	**100.0**±0.00
9	9.26 ±0.09	5.56 ±0.05	37.04 ±0.15	50.00 ±0.05	53.70 ±0.14	**86.90**±0.10	66.66 ±0.47	60.83 ±0.29	56.25 ±0.42	66.67 ±0.47	72.22 ±0.00	74.26 ±0.04	28.57	44.44	64.86	45.22	72.92±21.85
10	60.91 ±0.05	65.41 ±0.04	76.00 ±0.04	73.87 ±0.02	73.68 ±0.03	96.08 ±0.02	93.77 ±0.03	96.05 ±0.01	98.33 ±0.01	87.41 ±0.07	94.74 ±0.00	98.21 ±0.02	59.29	68.13	89.48	78.84	**99.11**±0.35
11	87.88 ±0.02	80.37 ±0.02	80.31 ±0.03	82.90 ±0.01	84.93 ±0.02	97.35 ±0.01	89.78 ±0.04	93.32 ±0.04	99.08 ±0.01	76.69 ±0.10	95.61 ±0.00	99.09 ±0.00	69.54	72.09	94.40	79.65	**99.58**±0.21
12	41.26 ±0.03	55.68 ±0.01	78.65 ±0.01	74.91 ±0.04	73.35 ±0.05	94.00 ±0.01	83.43 ±0.11	86.65 ±0.08	98.46 ±0.01	88.65 ±0.04	97.00 ±0.00	98.37 ±0.01	52.36	52.53	89.32	66.28	**99.03**±0.65
13	90.09 ±0.04	97.66 ±0.01	96.94 ±0.01	96.94 ±0.02	98.74 ±0.01	95.01 ±0.03	98.19 ±0.02	82.16 ±0.08	100.00 ±0.00	99.78 ±0.01	97.83 ±0.00	**99.80**±0.01	91.27	79.37	99.46	96.69	99.10±1.03
14	95.46 ±0.01	95.61 ±0.01	94.50 ±0.01	93.82 ±0.01	96.22 ±0.01	98.49 ±0.01	96.00 ±0.02	95.39 ±0.02	98.63 ±0.01	90.06 ±0.09	99.12 ±0.00	99.22 ±0.01	88.57	89.82	97.23	97.77	**99.93**±0.07
15	41.11 ±0.03	56.00 ±0.05	65.71 ±0.02	60.42 ±0.04	60.04 ±0.03	94.10 ±0.03	91.22 ± 0.04	90.96 ±0.13	99.24 ±0.01	88.21 ±0.04	92.80 ±0.00	97.86 ±0.01	37.76	43.97	68.71	70.11	**98.82**±1.71
16	79.37 ±0.03	84.92 ±0.02	82.54 ±0.04	91.27 ±0.05	90.87 ±0.02	93.57 ±0.05	70.90 ±0.39	94.73 ±0.04	95.63 ±0.06	98.53 ±0.02	**100.00**±0.00	95.93 ±0.06	44.26	78.01	91.67	88.27	98.90±2.45
OA	72.98 ±0.01	76.23 ±0.01	81.24 ±0.01	82.00 ±0.01	82.13 ±0.01	96.98 ±0.01	91.47 ±0.03	92.44 ±0.01	98.38 ±0.01	83.44 ±0.06	96.15 ±0.05	98.66 ±0.01	64.89	68.78	87.54	75.54	**99.28**±0.27
AA	59.41 ±0.01	65.23 ±0.02	75.98 ±0.01	77.36 ±0.02	79.53 ±0.01	94.96 ±0.01	94.14 ±0.01	91.19 ± 0.03	91.11 ±0.08	86.91 ±0.08	95.21 ±0.03	96.59 ±0.01	59.19	64.45	88.61	68.26	**97.04**±1.81
$\kappa(x100)$	68.62 ±0.01	72.66 ±0.01	78.58 ±0.01	79.41 ±0.01	79.54 ±0.01	96.55 ±0.01	90.20 ±0.03	91.37 ±0.01	98.15 ±0.01	80.82 ±0.07	95.60 ±0.03	98.48 ±0.01	60.11	64.47	85.93	72.10	**99.18**±0.00

TABLE V

OA, AA, AND κ VALUES ON THE KSC DATASET USING A FIXED SET OF 10% OF THE AVAILABLE LABELED SAMPLES (BOLD INDICATES THE BEST RESULTS)

Class	Classical Models					Deep Neural Networks							
	RF [88]	MLR [89]	GRU [90]	SVM [91]	LSTM [28]	ContextNet [82]	DPyResNet [76]	ResNet [75]	SSRN [83]	MS-3DNet [84]	ENL-FCN [85]	A^{2}S^{2}K-ResNet [48]	Free^2-ResNet
1	94.79 ± 0.01	95.57 ± 0.01	96.98 ± 0.01	95.43 ± 0.02	94.60 ± 0.00	99.78 ± 0.00	99.06 ± 0.01	94.73 ± 0.01	99.95 ± 0.00	96.42 ± 0.01	99.71 ± 0.00	99.95 ± 0.00	99.76±0.42
2	81.58 ± 0.05	88.74 ± 0.02	82.04 ± 0.02	83.71 ± 0.01	85.69 ± 0.02	98.79 ± 0.01	89.72 ± 0.03	66.45 ± 0.31	100.00 ± 0.00	95.88 ± 0.01	**100.00** ± 0.000	98.68 ± 0.02	99.92±0.19
3	86.09 ± 0.02	90.87 ± 0.05	89.13 ± 0.02	78.41 ± 0.22	91.16 ± 0.04	82.83 ± 0.05	81.84 ± 0.07	65.08 ± 0.19	99.66 ± 0.01	80.12 ± 0.17	**100.00** ± 0.00	98.72 ± 0.01	97.99±1.54
4	71.22 ± 0.06	42.00 ± 0.06	56.98 ± 0.06	27.17 ± 0.17	68.14 ± 0.05	78.41 ± 0.17	89.83 ± 0.04	73.62 ± 0.19	91.22 ± 0.05	90.06 ± 0.01	**98.67 ± 0.00**	94.27 ± 0.04	97.55±3.35
5	47.59 ± 0.06	25.06 ± 0.11	68.74 ± 0.09	22.99 ± 0.17	49.89 ± 0.11	74.22 ± 0.10	88.34 ± 0.10	60.74 ± 0.28	**100.00** ± 0.00	85.86 ± 0.03	98.61 ± 0.00	94.46 ± 0.05	97.70±2.70
6	48.22 ± 0.01	50.16 ± 0.03	63.27 ± 0.10	36.89 ± 0.08	54.21 ± 0.06	92.64 ± 0.05	88.54 ± 0.14	66.58 ± 0.32	98.45 ± 0.02	85.61 ± 0.03	**100.00** ± 0.00	99.82 ± 0.00	99.73±0.60
7	79.43 ± 0.10	75.53 ± 0.07	90.78 ± 0.03	87.94 ± 0.03	89.01 ± 0.01	94.40 ± 0.04	100.00 ± 0.00	66.20 ± 0.47	95.42 ± 0.05	90.75 ± 0.09	**100.00** ± 0.00	99.61 ± 0.01	99.21±0.88
8	78.61 ± 0.05	77.75 ± 0.02	90.03 ± 0.03	70.19 ± 0.07	92.53 ± 0.01	97.49 ± 0.01	94.81 ± 0.04	92.53 ± 0.08	99.80 ± 0.00	98.99 ± 0.01	**100.00** ± 0.00	100.00 ± 0.00	99.91±0.21
9	89.46 ± 0.01	89.32 ± 0.01	96.01 ± 0.02	85.33 ± 0.02	95.37 ± 0.03	99.92 ± 0.00	99.06 ± 0.00	90.82 ± 0.11	**100.00** ± 0.00	97.44 ± 0.03	**100.00** ± 0.00	**100.00** ± 0.00	99.80±0.44
10	88.43 ± 0.03	87.60 ± 0.01	91.09 ± 0.02	78.88 ± 0.07	94.03 ± 0.01	**100.00** ± 0.00	99.46 ± 0.00	87.59 ± 0.17	**100.00** ± 0.00	98.78 ± 0.01	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
11	95.58 ± 0.01	92.93 ± 0.02	96.02 ± 0.03	93.81 ± 0.01	96.11 ± 0.01	99.90 ± 0.00	99.90 ± 0.00	98.96 ± 0.02	**100.00** ± 0.00	98.67 ± 0.01	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
12	82.63 ± 0.03	93.60 ± 0.01	89.99 ± 0.04	86.98 ± 0.01	92.35 ± 0.04	99.17 ± 0.01	94.42 ± 0.06	84.26 ± 0.21	**100.00** ± 0.00	99.06 ± 0.01	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
13	99.60 ± 0.00	**100.00** ± 0.00	99.92 ± 0.00	**100.00** ± 0.00	99.96 ± 0.00	99.96 ± 0.00	99.96 ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
OA	86.17 ± 0.00	85.52 ± 0.00	90.13 ± 0.00	81.27 ± 0.01	90.18 ± 0.01	96.34 ± 0.01	95.61 ± 0.02	79.73 ± 0.21	99.29 ± 0.00	95.61 ± 0.02	99.25 ± 0.02	99.34 ± 0.00	**99.62**±0.17
AA	80.25 ± 0.00	77.62 ± 0.01	85.46 ± 0.01	72.90 ± 0.02	84.85 ± 0.01	93.65 ± 0.03	94.22 ± 0.02	80.58 ± 0.17	98.80 ± 0.01	93.66 ± 0.02	98.77 ± 0.02	98.88 ± 0.00	**99.35**±0.32
$\kappa(x100)$	84.59 ± 0.01	83.85 ± 0.01	88.99 ± 0.01	79.09 ± 0.01	89.06 ± 0.01	95.93 ± 0.02	95.11 ± 0.02	78.06 ± 0.22	99.21 ± 0.01	95.11 ± 0.02	99.13 ± 0.02	99.27 ± 0.00	**99.57**±0.00

the global and local attention modules, where the former is the top-attention module and the latter is included within residual blocks 1 and 2; the local attention net shares the same architecture that the proposed `Free`^{2}`-ResNet` includes only the local attention modules within residual blocks 1 and 2; the global attention net shares the same architecture that the proposed `Free`^{2}`-ResNet` includes only the global attention modules at the top of the network; and finally, the proposed `Free`^{2}`-ResNet` includes both the global and local attention modules.

Table III provides the obtained classification accuracy for different modules of the proposed framework. As shown in Table III, the results obtained by using the baseline framework (ResNet) with separate local and global attention mechanisms cannot achieve the results obtained by the proposed integrated framework. This, together with the classification results obtained by our method and the lower number of parameters required compared to other similar methods (see Table VII), positions our approach as a suitable alternative to other state-of-the-art HSI classifiers in terms of both accuracy and computational cost.

3) Separability of Classes in the Proposed Spectral–Spatial Feature Space: To study the separability of the different cover classes, t-SNE is used to visualize the test samples in the space formed by the considered spectral–spatial features, as shown in Fig. 5. Most test sample points in each category converge together for the IP, KSC, and UP datasets. The boundaries of different clusters are also clearly visible in the considered spectral–spatial space. It is not difficult to find that, although the separation degree between the test sample clusters of the IP and the KSC datasets is greater than that of the UP dataset, the clusters of the UP dataset are more pure (i.e., there are fewer misclassified samples). This means that the classification of the UP dataset is less disturbed by other class points. In contrast, the heavy mixing among the test sample clusters of the IP dataset affects the overall classification performance. The experimental results observed in Section III-D verify this point.

D. Classification Results

In this section, we provide the details of the classification results obtained for each dataset by using traditional and deep neural network classifiers over the classification benchmarks. Quantitative results are listed in Tables IV–VI, while Figs. 6–8 show the obtained classification maps.

IP: It should be noted that the IP scene suffers from high spectral mixing due to the low spatial resolution with which it was captured, making it one of the most complex HSI scenes to classify. From Table IV, it can be seen that the classification performance of traditional classifiers (RF, MLR, GRU, SVM, and LSTM) is not ideal, providing OA values below 85%. Furthermore, the classification accuracy of some classes is even lower than 10% for RF and MLR methods. Regarding transformers ViT, LeViT, and HiT, these models are computationally very complex and with a significantly larger set of parameters to be adjusted than traditional deep learning techniques, as it can be observed in Table VII.

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TABLE VI

OA, AA, AND κ VALUES ON THE UP DATASET USING A FIXED SET OF 10% OF THE AVAILABLE LABELED SAMPLES (BOLD INDICATES THE BEST RESULTS)

Class	Classical Models					Deep Neural Networks											
	RF [88]	MLR [89]	GRU [90]	SVM [91]	LSTM [28]	ContextNet [82]	DPyResNet [76]	ResNet [75]	SSRN [83]	MS-3DNet [84]	ENL-FCN [85]	A^{2}S^{2}K-ResNet [48]	ViT [86]	LeViT [87]	HiT [58]	SSAN [55]	Free^2-ResNet
1	91.11 ± 0.01	92.30 ± 0.00	93.34 ± 0.00	94.30 ± 0.01	95.47 ± 0.01	99.56 ± 0.00	98.35 ± 0.01	96.82 ± 0.02	99.85 ± 0.00	99.36 ± 0.00	**99.98** ± 0.00	99.91 ± 0.00	90.20	93.62	96.19	84.81	99.87±0.14
2	98.11 ± 0.01	96.18 ± 0.00	97.54 ± 0.00	97.65 ± 0.00	96.91 ± 0.00	99.85 ± 0.00	98.76 ± 0.01	98.59 ± 0.01	99.98 ± 0.00	99.80 ± 0.00	**100.00**± 0.00	99.99 ± 0.00	89.99	91.42	92.79	97.85	**100.0**±0.00
3	67.71 ± 0.01	72.75 ± 0.01	77.08 ± 0.04	81.26 ± 0.02	78.01 ± 0.01	99.19 ± 0.00	94.22 ± 0.03	90.01 ± 0.06	99.68 ± 0.00	98.02 ± 0.02	99.68 ± 0.00	**99.88**± 0.00	71.59	84.27	93.21	26.82	99.00±0.50
4	88.20 ± 0.01	89.28 ± 0.00	93.22 ± 0.02	94.63 ± 0.01	94.92 ± 0.01	99.80 ± 0.00	99.20 ± 0.01	99.32 ± 0.00	99.92 ± 0.00	99.71 ± 0.00	98.94 ± 0.00	**99.95**± 0.00	96.06	96.98	97.33	98.78	99.84±0.13
5	98.93 ± 0.00	99.42 ± 0.00	99.42 ± 0.01	99.20 ± 0.00	99.26 ± 0.00	99.91 ± 0.00	99.72 ± 0.00	99.81 ± 0.00	99.94 ± 0.00	99.94 ± 0.00	**100.00**± 0.00	**100.00**± 0.00	99.42	99.93	99.96	99.85	**100.0**±0.00
6	72.14 ± 0.02	77.45 ± 0.01	87.41 ± 0.02	90.58 ± 0.01	87.85 ± 0.01	99.75 ± 0.00	98.52 ± 0.01	99.41 ± 0.00	99.95 ± 0.00	99.43 ± 0.00	99.87 ± 0.00	99.91 ± 0.00	90.32	95.97	99.91	38.36	**100.0**±0.00
7	75.69 ± 0.02	55.69 ± 0.04	85.38 ± 0.04	85.71 ± 0.01	80.23 ± 0.01	98.37 ± 0.02	97.37 ± 0.01	96.90 ± 0.02	**100.00**± 0.000	99.18 ± 0.01	**100.00**± 0.00	**100.00**± 0.00	81.23	93.01	98.22	47.12	99.86±0.15
8	89.64 ± 0.01	87.04 ± 0.01	88.56 ± 0.02	88.20 ± 0.00	88.49 ± 0.01	98.48 ± 0.01	84.51 ± 0.07	92.00 ± 0.04	98.28 ± 0.02	97.13 ± 0.01	99.69 ± 0.00	98.88 ± 0.01	90.02	95.30	99.15	90.38	**99.80±0.18**
9	99.77 ± 0.00	99.77 ± 0.00	99.84 ± 0.00	99.84 ± 0.00	99.88 ± 0.00	99.26 ± 0.01	99.60 ± 0.00	98.88 ± 0.01	99.39 ± 0.00	99.74 ± 0.00	**100.00** ± 0.00	99.78 ± 0.01	98.01	99.79	99.77	98.82	99.96±0.06
OA	90.41 ± 0.01	89.87 ± 0.00	93.34 ± 0.00	94.19 ± 0.00	93.45 ± 0.00	99.57 ± 0.00	97.05 ± 0.01	97.38 ± 0.01	99.77 ± 0.00	99.35 ± 0.00	99.76 ± 0.00	99.85 ± 0.00	86.41	89.68	92.00	83.28	**99.90**±0.03
AA	86.81 ± 0.01	85.54 ± 0.01	91.31 ± 0.01	92.38 ± 0.00	91.23 ± 0.00	99.35 ± 0.00	96.69 ± 0.01	96.86 ± 0.01	99.66 ± 0.00	99.15 ± 0.00	99.70 ± 0.00	99.81 ± 0.00	89.64	94.98	97.38	75.86	**99.82**±0.05
$\kappa(x100)$	87.10 ± 0.00	86.46 ± 0.00	91.15 ± 0.00	92.29 ± 0.00	91.30 ± 0.00	99.43 ± 0.00	96.08 ± 0.01	96.52 ± 0.01	99.69 ± 0.00	99.13 ± 0.00	99.72 ± 0.00	99.81 ± 0.00	82.60	86.83	89.77	77.19	**99.86**±0.00

TABLE VII

PARAMETERS AND RUNTIME (IN SECONDS) REQUIRED BY DIFFERENT METHODS

Model	**Parameters**	**Runtime** (s)
ContextNet [82]	554.1K	1530
DPyResNet [76]	22.3M	442
ResNet [75]	21.9M	500
SSRN [83]	364.1K	571
MS-3DNet [84]	269K	1520
ENL-FCN [85]	113.9K	54
3DCNN	1516.5K	125
A3DCNN	1516.5K	142
ViT [86]	2600.0K	63
LeViT [87]	16940.0K	149
HiT [58]	51180.0K	112
SSAN [55]	2836.05K	2383
A^{2}S^{2}K-ResNet [48]	370.8K	620
Free^2-ResNet	365.8K	272

Consequently, they suffer a drop in accuracy due to the low number of training samples to cover the high data variability, which causes the models to overfit. Also, the SSAN struggles when classifying IP scene, as this model is based on a much larger input window (27 × 27) than the one used in the proposed model (9 × 9). This means that the parameters cannot be conveniently adjusted. In contrast, the other deep neural network classifiers achieve outstanding results (in some way due to the use of spatial information provided by neighboring pixels), surpassing the 90% OA. In particular, the accuracy of the ENL-FCN and the proposed `Free`^{2}`-ResNet` even reach a 100% for some categories. In most cases, the proposed `Free`^{2}`-ResNet` classifier obtains the best performance among all the methods, followed by the A^{2}S^{2}K-ResNet and SSRN.

As for the classification maps, intuitively, the classification maps of SSRN, A^{2}S^{2}K-ResNet, and the proposed `Free`^{2}`-ResNet` classifiers are more accurate than those provided by the other deep neural network classifiers (see Fig. 6). This suggests the effectiveness of including spatial information in these three methods. Considering the quantitative and qualitative results together, the classification performance of the proposed `Free`^{2}`-ResNet` classifier is superior to that achieved by the other tested methods.

KSC: Captured by the same sensor as IP, this scene is characterized by the low number of labeled data, posing a challenge to deep models that generally tend to overfit the training in the absence of enough training samples to cover all the data variability. In Table V, the OA of the best traditional classifier is over 85%; in particular, the two recurrent networks GRU and LSTM achieve a remarkable performance, which represents an improvement with regard to the results for the IP dataset. Among deep neural network classifiers, the classification performance of SSRN, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed `Free`^{2}`-ResNet` is better than that of the other methods, with many categories classified perfectly. In particular, the OA of SSRN, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed `Free`^{2}`-ResNet` is over 99 %. Nevertheless, the proposed `Free`^{2}`-ResNet` exhibits the best OA, AA, and κ values, proving the consistency and robustness of the classification.

Regarding the classification maps of Fig. 7, it can be seen that the map of the proposed `Free`^{2}`-ResNet` method is the most similar to the ground truth. In the other classification maps, some pixels of the *Graminoid* class are classified incorrectly, as it can be seen in the top-right corner of the classification maps. A possible reason is that spatial features help to improve the classification performance of the proposed method.

UP: Compared to the IP and the KSC dataset, the UP dataset contains artificial land cover types and the spatial resolution is higher, so the spectral differences are more apparent and the spatial features are more useful. From Table VI, we can see that both classical and deep neural network classifiers achieve better classification performance with UP than with the other two datasets. The OA of classical methods exceeded 90% (except for the MLR, which is close to 90%), while the OA of deep neural network classifiers exceeded 99% (except for the DPyResNet and the ResNet). This means that the separability among categories and the abundant spatial information does have a beneficial impact on the improvement of classification results. For most categories, the classification accuracy of ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed `Free`^{2}`-ResNet` is similar and better than that of the other methods. Nevertheless, the proposed `Free`^{2}`-ResNet` still outperforms them, providing the best OA, AA, and κ values. Focusing on transformers, ViT, LeViT, and HiT still present overfitting problems due to the complexity of their architectures. It is easily observable in Table VII that the order of magnitude of the parameters is significantly higher than that of the other neural networks. Once more, SSAN is negatively affected by the spatial size of the input data.

As shown in Fig. 8, the classification maps of SSRN, ENL-FCN, A^{2}S^{2}K-ResNet, and the proposed `Free`^{2}`-ResNet` are

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TABLE VIII

OA (%) OF DPYRESNET, RESNET, SSRN, MS-3DNET, ENL-FCN, A^{2}S^{2}K-RESNET, AND THE PROPOSED FREE^2-RESNET USING DIFFERENT SPATIAL WINDOW SIZES AND 10% OF THE AVAILABLE LABELED SAMPLES (BOLD INDICATES THE BEST RESULTS)

Data sets	Spatial Window	Models						
		DPyResNet [82]	ResNet [75]	SSRN [83]	MS-3DNet [84]	ENL-FCN [85]	A^{2}S^{2}K-ResNet [48]	Free^2-ResNet
IP	7×7	91.31±0.006	87.47±0.008	98.10±0.005	77.33±0.015	69.25±0.001	98.77±0.001	**99.06±0.2365**
KSC		82.12±0.033	76.03±0.052	99.06±0.004	95.95±0.007	27.40±0.001	**99.48±0.003**	99.44±0.1975
UP		96.30±0.006	96.33±0.001	99.54±0.002	98.26±0.001	90.32±0.001	99.72±0.001	**99.84±0.062**
IP	9×9	91.47±0.029	92.44±0.006	98.38±0.004	83.44±0.060	96.15±0.054	98.66±0.004	**99.28±0.2705**
KSC		95.61±0.019	95.61±0.019	99.29±0.004	95.61±0.019	99.25±0.020	99.34±0.001	**99.62±0.1684**
UP		97.05±0.010	97.38±0.007	99.77±0.001	99.35±0.001	99.76±0.002	99.85±0.001	**99.90±0.0287**
IP	11×11	92.66±0.003	92.67±0.008	98.18±0.003	68.90±0.150	56.48±0.001	98.55±0.002	**98.71±0.5693**
KSC		95.47±0.007	93.36±0.023	98.83±0.006	94.27±0.002	53.44±0.001	99.47±0.003	**99.48±0.1212**
UP		97.21±0.016	97.83±0.006	99.82±0.001	99.45±0.001	93.92±0.001	99.90±0.000	**99.93±0.0394**

TABLE IX

ACCURACY PER CLASS, OA, AA, AND κ VALUES FOR IP AND UP DATASETS USING A FIXED SET OF 20 OF LABELED SAMPLES PER CLASS (BOLD INDICATES THE BEST RESULTS)

Class	Indian Pines (IP)			University of Pavia (UP)		
	D-SSN [53]	S3Net [53]	Free^2-ResNet	D-SSN [53]	S3Net [53]	Free^2-ResNet
1	**100.00±0.00**	**100.00±0.00**	**100.0±0.00**	94.67±3.40	95.27±2.59	**95.39±2.99**
2	85.92±5.69	**88.96±5.87**	86.85±2.68	92.48±6.46	95.77±4.03	**97.90±1.39**
3	88.21±5.24	89.89±2.87	**90.96±1.27**	89.85±6.23	90.40±6.35	**93.44±4.18**
4	98.11±1.93	**99.49±1.08**	99.36±0.69	94.16±4.19	**96.05±3.20**	95.03±2.91
5	**92.74±3.97**	92.44±4.05	90.15±2.72	**99.98±0.05**	99.86±0.13	99.56±0.52
6	96.65±3.43	96.63±2.69	**98.31±1.07**	95.80±3.84	**99.21±1.35**	98.12±1.40
7	**100.00±0.00**	**100.00±0.00**	**100.0±0.00**	99.21±0.79	**99.92±0.25**	99.63±0.31
8	**100.00±0.00**	**100.00±0.00**	**100.0±0.00**	92.09±5.49	**96.90±3.77**	95.79±1.63
9	**100.00±0.00**	**100.00±0.00**	**100.0±0.00**	**99.77±0.24**	98.83±0.88	98.99±0.36
10	87.79±3.94	88.58±4.82	**89.60±3.87**	-	-	-
11	81.36±5.85	85.03±5.00	**90.37±1.98**	-	-	-
12	**94.38±2.59**	92.93±5.08	90.23±3.52	-	-	-
13	99.30±1.11	99.14±1.61	**99.89±0.22**	-	-	-
14	97.95±2.23	98.02±3.23	**99.71±0.27**	-	-	-
15	95.22±4.57	**99.07±1.05**	96.12±3.87	-	-	-
16	**99.18±1.10**	99.04±1.23	**99.18±1.10**	-	-	-
OA	89.91±1.67	91.54±1.13	**92.71±0.37**	93.77±2.69	96.27±1.98	**97.06±1.05**
AA	94.80±0.83	95.58±0.50	**95.67±0.36**	95.34±1.12	96.91±0.90	**97.09±0.94**
$\kappa(x100)$	88.54±1.88	90.38±1.27	**91.68±4.23**	91.87±3.37	95.11±2.55	**96.11±1.39**

most similar to the ground truth. A careful observation of the results reveals that the areas containing bricks in the classification maps of SSRN and ENL-FCN are mixed. The proposed Free^2-ResNet still appears to be the best choice for HSI classification among all considered methods.

In summary, and regardless of the dataset, the classification performance of deep neural network classifiers considering both spectral and spatial information is higher than that achieved by traditional classifiers using only spectral features. This is because, on the one hand, deep neural networks can mine more abstract features to separate different categories. On the other hand, spatial features can help to maintain the purity of intraclass features and the delineation of edges in local areas. Among the deep neural network classifiers considered in our experiments, the performance of the A^{2}S^{2}K-ResNet and the proposed Free^2-ResNet is higher than that of the other classifiers. The OA, AA, and κ values of the proposed Free^2-ResNet are the best among all the methods for the three considered datasets, with its OA exceeding 99%. By the same token, classification maps of the proposed Free^2-ResNet are also the most similar with regard to the corresponding ground truth maps. It can be inferred that the proposed Free^2-ResNet method is the most suitable one for diverse practical applications.

E. Discussion

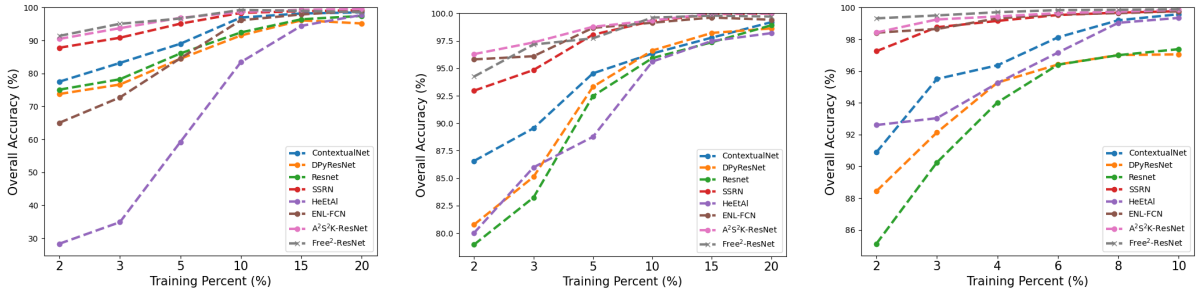
By effectively combining spectral and spatial features, the proposed Free^2-ResNet method obtains ideal classification performance with a parameter-free attention network that does not compel the computational load of the model. In order to analyze the effectiveness and stability of Free^2-ResNet classifier in detail, this section analyzes three aspects.

1) Influence of the Spatial Window Size: Our method uses a spatial window in a manner similar to the other six deep neural network models that have been considered for comparative purposes, i.e., DPyResNet, ResNet, SSRN, MS-3DNet, ENL-FCN, and A^{2}S^{2}K-ResNet. In this context, when classifying a spectral pixel, the model exploits the spatial information provided by the surrounding neighbors. The impact of these neighbors is calculated with the objective of selecting an appropriate and fair window size that does not unfairly bias the classification result in an unnaturally optimistic way.

Table VIII reports the OA of the seven methods as a function of different spatial window sizes (7 × 7, 9 × 9, and 11 × 11), considering 10% of the available labeled samples for training the classifiers with the three considered datasets. The best results are highlighted in bold. It can be seen that the proposed Free^2-ResNet classifier achieves in most of the cases the best OA value for the different datasets and spatial window sizes among all methods, with the only exception of KSC 7 × 7, where A^{2}S^{2}K-ResNet provides the best result, closely followed by the proposed method. Furthermore, focusing on the same dataset with different spatial sizes, the OA of the A^{2}S^{2}K-ResNet and the proposed Free^2-ResNet vary very slightly, as opposed to the results yielded by the other five deep classification methods, which are clearly affected by the spatial size of the neighborhood window. This clearly indicates that the stability of A^{2}S^{2}K-ResNet and Free^2-ResNet is higher. In practice, effectiveness and stability are equally important. Therefore, the proposed Free^2-ResNet classifier appears to provide a better tradeoff than other methods.

2) Influence of the Training Sample Size: In classification tasks, the acquisition of labeled samples is often labor-intensive and costly. How to achieve ideal classification with a limited number of samples is an important problem to be addressed and a hot topic within the scientific community of HSI processing. Here, we test the classification performance of all deep neural network classifiers using 2%, 3%, 5%, 10%, 15%, and 20% of the available labeled samples for training. The experimental results obtained for the three considered datasets are shown in Fig. 9. Further details are provided as follows.

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(a) (b) (c)

Fig. 9. OA achieved by different methods with varying training sample sizes: (a) IP, (b) KSC, and (c) UP.

Focusing on the IP scene, in Fig. 9(a), it can be seen that, with the increase of the training percentage, the OA of the proposed Free^2−ResNet method is always higher than that of the other methods (with a slowly rising trend), followed by the A^{2}S^{2}K-ResNet and the SSRN methods. Regarding KSC scene, in Fig. 9(b), it can be seen that the OA of the proposed Free^2−ResNet is slightly lower than that of A^{2}S^{2}K-ResNet when the training samples percentage is between 2% and 5%. The possible reason is that the number of the labeled samples of KSC is too small so that the classification result may be unstable when the training percent is too small. Just as shown in Fig. 9(b), the OA of the proposed method for 3% training percent is better than that for 2% and 5%. At the same time, random sampling for this dataset can also have an impact on the results of the algorithm when the training percent is small. However, these problems have been alleviated as the training percentage increased. When the training percent is equal or larger than 10%, the accuracy of the proposed Free^2−ResNet method exceeds the A^{2}S^{2}K-ResNet and becomes the best one. Finally, regarding UP scene, in Fig. 9(c), it can be seen that the proposed Free^2−ResNet steadily maintains higher accuracy than other methods with the increase of the number of training samples, followed by the A^{2}S^{2}K-ResNet, the ENL-FCN, and the SSRN methods. In other words, the proposed method is able to obtain better classification accuracy compared to other methods with fewer training samples. This demonstrates the robust accuracy and stability of the proposed Free^2−ResNet classifier, which is fully applicable to scenarios with limited training samples.

3) Influence of Class Imbalance Training: The issue of class imbalance arises with a substantial disparity in the quantity of available training samples across different classes. This issue leads to classifiers being predisposed toward the majority class, which consequently results in poor classification performance on the minority classes. In the context of hyperspectral scenes, the challenge of class imbalance is particularly significant. This is due to the presence of specific classes in a limited number of pixels, as well as the high degree of similarity that exists between some classes within the scene. In the case of extreme class imbalance, this may tend to predict the majority class and not learn the underlying characteristics of the minority classes, leading to weak generalization performance and adversely affecting classification. To overcome this challenge, it is essential to address the class imbalance issue by balancing the class distribution, for instance, by determining a fixed number of training samples per class.

This experiment involved a comparison of the S3Net and D-SSN approaches with the proposed Free^2−ResNet for two distinct scenes, IP and UP. S3Net employs differentiated input and adaptive cross-entropy loss, while D-SSN is constructed by two Siamese networks without differentiated input, adaptive cross-entropy loss, and with the addition of weighted contrastive loss. The inclusion of the last draws attention to the complex features, with the aim of increasing their representation, such as imbalance data. The results included in Table IX indicate that the proposed Free^2−ResNet method yields improved performance in terms of OA, AA, and $\kappa(x100)$, providing valuable insights into its advantages over the previous experiment. Considering that this experiment involved 20 labeled samples per class, the training stage for IP and UP ensures 3.12% and 0.42% of samples, respectively.

IV. Conclusion

In this article, we develop a new attention network for HSI classification that provides an excellent tradeoff between the obtained classification accuracy and the number of parameters required. A relevant property of our attention module is that, in contrast to existing channelwise and spatialwise attention modules, it infers 3-D weights for the feature map in a layer without adding parameters to the original network. Our experimental results with three widely used benchmark HSIs demonstrate that the proposed approach outperforms other state-of-the-art classifiers in terms of both accuracy and computational complexity.

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